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ANTH 591

Final Report

Human-AI Interaction Pilot Study Report

Introduction

The use of generative artificial intelligence tools has grown dramatically since the introduction of OpenAI's ChatGPT in 2022, followed by Anthropic's Claude, Google's Gemini, DeepSeek, and Meta's llama AI. The rapid introduction and implementation of these tools has forced institutions, industries and individuals alike to contend with technology that is said to simultaneously be both transformative and highly overrated. Public discourse surrounding the use of AI tools has oscillated between fears of both job loss and skepticism about the efficacy of AI tools in settings where its adoption is being encouraged...or even threatened. Despite the volume of this public discourse, what we know about AI implementation and its use is still beginning to be understood. Since AI's viral injection into the world at large, there have been widely circulated misleading articles claiming that GPT-4 passed the bar exam (Arredondo 2023). While other credible research conducted by MIT suggests that AI use may contribute to cognitive decline in humans (Kosmyna et al. 2025). At this stage, this research has shaped public perception of AI without producing much of a clear picture of how students are integrating these tools into their academic lives, or where they are learning to use them. With a promise for AI-driven force multiplied productivity on one side and a risk of human cognitive decline on the other, this gap in understanding carries real consequence for academic outcomes and the integrity of higher education.

Research Questions

- How do students learn to use AI tools?

There is a gap in research and understanding of the ways in which students currently interact with AI tools. Given that most schools are not explicitly teaching students how to interact with AI tools, they learn through unconventional sources.

- How do students interact with AI tools?

Most instances of AI interaction within academia happens out of view of educators and academic spaces as AI use is generally frowned upon. Given that AI tools are shaping academic outcomes, it is important to understand to what extent the agency of an AI tool is having over a students work

- How successful are AI tools at the tasks given to them by students?

Simply put, how helpful is AI for a student? Do students recognize when an AI tool is hallucinating? Or possibly taking them down the wrong path?

Research Design

Variables & Measurement

How students learn to use AI tools: The purpose of this variable was to identify sources of knowledge, processes, and overall impact of information gathered from non-traditional spaces. This variable was measured with semi-structured interviews aimed at identifying sources, and methods used to apply AI in academic work.

How does AI respond to user interaction: The purpose of this variable was to identify how AI is responding to user input and effecting change on the users' workflow and perception of specific topics while using AI tools. This variable was measured using diary studies methods in the form of screenshot transcripts of each Human-AI interaction.

How students interpret and respond to AI: The purpose of this variable was to identify patterns of interaction between students and AI tools. This variable was measured with diary studies of AI tool use to identify what prompts, outputs and revisions are used to achieve a desired outcome. Data collected came in the form of an audio recording of their experiences with each test session conducted. Audio recordings were then converted into transcripts for analysis.

Sampling Methods

For this pilot study participants were selected based on their proximity to my own social circle to assure that research and data was conducted and collected in the short window provided. Of those within my social circle I chose to select participants that came from different academic backgrounds to observe differences in AI usage across academic fields.

Data Collection Methods

- Semi-Structured Interviews: This data collection method was used to inform the participant of the structure of the study, build rapport and gather information about their background with AI tools, opinions, and experiences with technology. This method was conducted online via zoom.
- Diary Studies: This data collection method was used to understand how participants were interacting with AI, how they periceived AI interactions and how AI was interacting with the participants. By observing both the AI tool interaction and correlating that with the participants accounting of the interaction I was able to observe the interactions asynchronously without risk of effecting the participants behavior with my presence.

Data Analysis & Tools

- MaxQDA Software: For the initial coding phase of data analysis, I chose to utilize the qualitative analysis software known as MaxQDA. This software allowed for me to neatly organize, code and compile collected data for further analysis. This software afforded me the ability to streamline my workflow and create detailed reports on the codes and observed behavioral patterns between human and AI.
- Ethogram Inspired Analysis: As a secondary phase of data analysis, an ethogram inspired analysis affords this pilot study with a more detailed and consolidated view of behavioral patterns within Human-AI interactions. Behaviors are categorized and described based on the context in which they were observed within both Human-AI interaction transcripts and correlated descriptions observed in participant diary transcripts.

Time Frame For Project

Task	**Academic Week**
AI Requirements Analysis	Week 3
Research Subject Sampling	Week 4
Semi-Structured Interviews	Week 5-6
Diary Study Implementation	Week 7-8
Data Analysis	Week 9-10
Report	Week 10

Actual Research Activities

According to my original research plan, I was able to successfully create a list of requirements to inform the sample of participants that I pursued for this research project and I was able to identify individuals that met my base requirements based on that initial analysis. I was able to successfully conduct semi-structured interviews with participants both before and after the main data collection phase.

Initially my research design for this pilot study focused on user experience (UX), observed AI use behaviors and AI policy. I had planned participant observations of test participants using AI to make connections between observed behaviors, AI tool use, project/assignment outcomes and how AI policy may play a part in said observed behaviors. Ultimately, I felt that a primary focus on UX and UI of AI tools as sole drivers of user behavior would do more to benefit the usability of AI tools rather than effect needed innovation in classroom curriculum based on my observations. So rather than focusing on a heuristic analysis of user behavior, I opted to focus on an ethogram inspired approach that focused more on defining identified behavioral patterns between both Humans and AI. This subtle shift in focus aligns more with what my larger thesis research seeks to focus on identifying behavioral patterns between students and AI to find solutions for innovating curriculum in a world with AI. In place of in-person participant observation, I opted to use diary study methods that allowed each participant to recount their experiences with AI tools for each planned session. This allowed each participant to complete each data collection session on their own time without having to set aside a specific amount of time with me and a specific location. I feel that this approach also allowed each participant to feel more comfortable to interact with AI in a far more natural state, I feared that my presence would effect the outcome of data collected.

Initially I had also planned to utilize surveys for participant demographic information but opted out of that and gathered this information in the semi-structured interview phase of research. I decided to do this primarily because it became obvious that asking participants to complete additional surveys without some sort of compensation was going to be difficult. While I was able to successfully complete pre-study semi-structured interviews with all participants, I was only able to complete a semi-structured exit interview with participant 2.

<u>Description of Data Collected</u>

AI Use Assessment Guidelines:

The guidelines listed below are what guided participants through each interaction with their AI tool of choice.

Session 1: Information Seeking Apply the following to a class assignment you're working on during the week. The purpose of this session is to observe how you're able to gather information from an AI, so feel free to ask these questions in whatever way works best for you. • Ask the AI to explain a concept that is complex or unfamiliar to you. • Ask follow-up questions that are genuine to the conversation. • Ask the same question in two different ways to see if you get a better answer. • Describe the experience in a video or voice memo and email or text to Austyn. • Screenshot or download a transcript of the full AI interaction and email to Austyn.
Session 2: Problem Solving/Decision Support Ask the AI to generate ideas/solutions to a specific assignment you have. For example, ideas on research topic for a final assignment within a given topic of study. The purpose of this session is to observe how you're able to problem solve, negotiate ideas and collaborate with AI tools. • Ask the AI to generate options in the form of research topics, solutions to a problem.

• Ask for step by step instruction to achieve your goals (Outline for a paper, pseudocode for a coding project, etc.) • Ask the AI for its opinion on the options it has provided based on the information you have provided it with as well • Describe the experience in a video or voice memo and email or text to Austyn. • Screenshot or download a transcript of the full AI interaction and email to Austyn.
Session 3: Process Comparison Complete an assignment without AI assistance. Once you have completed said assignment have AI complete the same assignment from start to finish. The purpose of this assignment is to compare the results of your own process versus the process of having AI complete an assignment for you. In your video/voice memo diary, make note of how the processes differed from one another. • Choose a straight forward assignment, likely a research paper or a coding assignment (I am open to whatever though, it would be interesting to see what AI can and can't do). Feel free to use assignments you have already completed as long as you remember doing it. • After you have completed the assignment on your own, reflect on your process for completing said process and describe the experience in a video or voice memo and email or text to Austyn. • Now, have AI complete the same assignment. Utilize whatever methods of prompt engineering work best for you.

Requirements Analysis:

The purpose of compiling a list of requirements for AI use was to help with my sampling strategy in identifying qualified participants as it pertains to using AI. Given that my main method of study for this pilot study was diary studies, I needed to identify participants who would be able to collect and compile data with as little supervision as possible.

Functional Requirements	**Description**
Computer w/Mouse, Keyboard & Monitor	Participant must have access to a computer in order to access AI tools
Internet Connection	Internet is required for most AI tool
Ability to read and write	Participant must be able to read and write to interact with AI
Access to email to login to AI tool	Email account is required to access AI tools
Basic technological literacy	Participant must have the knowledge required to record audio logs for diary studies methods and be able to take screenshots of their AI tool interactions, compile said interactions and recordings and email them to me.
AI Tool Requirements	- AI tool that participant uses must be easily accessible and common within the ecosystem of AI technology - AI tool must allow for screen shots to be taken or be able to easily download copy of AI interaction transcript

Semi-Structured Interview:

Data collected was used to gather more personal insight about the backgrounds of participants, mainly their field of study, stage in their academic career, experience with AI and their interest in AI. Here I aimed to build rapport with each participant so that they felt more comfortable with me analyzing their use of AI.

18 I have some questions set aside, but I kind of want to just discuss a little bit here. Actually, don't know. We'll jump into some questions that I have. ~ This is kind of ~ semi-structured, but just kind of relax. ~

19 So kind of just going from the start of like your experience with AI, ~ why did you first start using AI in the first place?

20 speaker-1 (01:25)
21 The very first time that I used AI at all was actually for a class where I was taking an evolutionary biology class. I dont remember what the exact course title was, but it was basically like an introduction to evolutionary biology. And that was right when GPT-4 first launched. And the final for that class had you write a final exam.

22 So your final exam was writing a final exam, and then you took someone else's final exam. ~ And part of the prompt for that was to feed it into GBT-4 to see if it was easily solvable by an AI and if it was to rewrite it until it wasn't, So do something that will require you to have actually attended the class and

23 speaker-0 (01:59)
24 Okay.

25 okay.

26 speaker-1 (02:21)
27 And that was when it couldn't do image processing. So one of the really common techniques was just like, instead of have it be text, have it be like a JPEG, which totally wouldn't work now, but did then.

28 speaker-0 (02:35)
29 This would have been like 2020.

30 speaker-1 (02:37)
31 I think 2021. And I had just done a project where I wrote a bunch of k-means clustering code in R at the same quarter, the quarter before, something like that, where I was doing some geochemical sourcing for ceramics and trying to cluster out specific ceramic shards based on their geochemical origin.

32 speaker-0 (02:39)
33 Yeah.

34 speaker-1 (03:05)
35 And so had just written a bunch of k-means clustering and I was like, wow, this took me like months to figure out how to do like what k-means clustering is. I was like, you know, I've heard it's good at writing code. So let me see if I can like tell it to write me a k-means clustering algorithm that's based on, you know, these parameters that I know I need it to have. I was like, huh, okay. Well, it wrote that in like 30 minutes. ~ That's good to know. So that was like the very first time that I used that.

36 And then I didn't find it particularly helpful for most things that I do. I'm not interested in it replacing my ability to do things. And so if it's a question of do I want it to do something for me or do I want to figure out what I need to do for myself, generally I'll choose the latter. And then I started really intensively looking at

- The image above shows a snippet of the exit interview conducted with participant 2. Here we are discussing their experience with AI, and how they came to understand the technology outside of academia.

Participant	Academic Stage	Field of Study	AI Experience	Interest in AI
Participant 1	Undergraduate (Senior)	Engineering	Novice	Interested in the ways it can help them in their career but only understands AI from a base level. Has some coding experience
Participant 2	Graduate	Anthropology	Experienced	Understanding AI seems to be a personal interest, as well as something that they hope to implement into their own research in an ethical data analysis way. Has

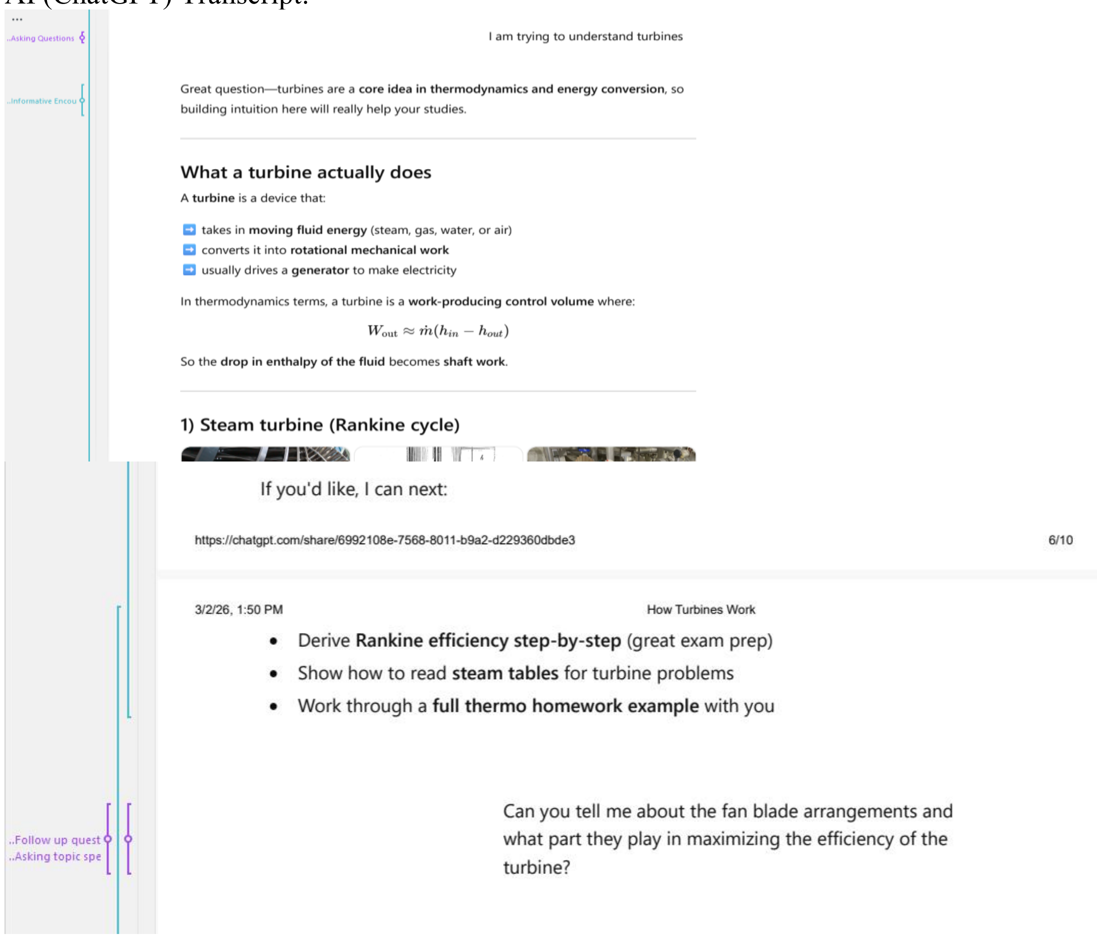
				extensive coding experience

Diary Study Data:

Data collected here was used to identify both use behavior and AI output behavior. Data collected came in the form of audio transcripts and screenshots/transcripts of each interaction.

Session 1: Examples are from session 1 of data collection phase.

- Participant 1:
    - AI (ChatGPT) Transcript:



..Asking Questions

I am trying to understand turbines

..Informative Encou

Great question—turbines are a **core idea in thermodynamics and energy conversion**, so building intuition here will really help your studies.

What a turbine actually does

A **turbine** is a device that:

- takes in **moving fluid energy** (steam, gas, water, or air)
- converts it into **rotational mechanical work**
- usually drives a **generator** to make electricity

In thermodynamics terms, a turbine is a **work-producing control volume** where:

$$W_{out} \approx \dot{m}(h_{in} - h_{out})$$

So the **drop in enthalpy of the fluid** becomes **shaft work**.

1) Steam turbine (Rankine cycle)

If you'd like, I can next:

https://chatgpt.com/share/6992108e-7568-8011-b9a2-d229360dbde3 6/10

3/2/26, 1:50 PM How Turbines Work

- Derive **Rankine efficiency step-by-step** (great exam prep)
- Show how to read **steam tables** for turbine problems
- Work through a **full thermo homework example** with you

..Follow up quest

..Asking topic spe

Can you tell me about the fan blade arrangements and what part they play in maximizing the efficiency of the turbine?

o Diary Transcript:

...	1	Other (00:01.182)
..sufficient interactior	2	Session one for AI use assessment. Excuse me. The experience of trying to learn a generally new topic from an AI was overall pretty good. I fear that I slightly misinterpreted the request and it was not directly related to an assignment that I have.
..Explaining context o	3	More just to a general topic that aligns with one of my classes. I asked the AI to tell me about Sorry A Turbine. I want to know about turbines. I am in thermodynamics right now and turbines are coming up as there are many thermodynamic principles involved and In my studies, I realized that I didn't quite know how a turbine worked. So I asked the AI
..sufficient interac ..AI use literacy ..sufficient interac ..User error reflect	4	I'd say it did a pretty good job. It's, they definitely do a lot of overviews. think that's kind of one of the hard things about using AI is that you really need to know what to ask it. It is not likely to, you know, curate like a comprehensive course for you or anything. It's entirely possible that if I asked for a very detailed workup, it would give me that.
..Prompt use compa ..sufficient interacti	5	There is of course a lot of higher concepts involved in that sort of thing as well as mathematics and so on. The differences that I got out of asking, I asked the same question twice. I literally asked it to explain what a turbine was just in a slightly different way. And it basically did the same thing just in a slightly different way.

- Participant 2:
  - o AI (Llama Gemma 3B & CoPilot) Transcript:

..Information seeking

I have a GamaKay TK75 that is not registering either windows key. What is the appropriate fix, and are there any security concerns associated? What do alternative options look like?

edit

..Demonstrates under

Okay, let's break down the issues you're experiencing with your GamaKay TK75 and explore potential fixes, security implications, and alternative options. This is a common issue with older keyboards and can be frustrating.

Understanding the Problem: Windows Key Failure

The GamaKay TK75 (and similar keyboards) often rely on microswitches that are prone to wear and tear, particularly the Windows keys. These keys are used frequently, leading to eventual failure. The symptoms you describe – not registering the Windows keys – strongly suggest a mechanical failure within the key switch itself. It's almost always a hardware problem, not a software one.

Appropriate Fixes (Ordered by Recommended Approach)

1. Keycap Cleaning & Lubrication (First Try - Least Invasive):
   - What it does: Dirt, grime, and old lubricant can interfere with the switch's movement. Cleaning and relubricating the switches can sometimes restore function.
   - How to do it:
   - Remove Keycaps: Use a keycap puller (recommended – don't pry) to remove the Windows keys.
   - Clean: Wipe the keycaps and switches with isopropyl alcohol (90% or higher) and a microfiber cloth.

o Diary Transcript:

...	1	Other (00:03.246)
..Explaining context o	2	Hi Austin, I'm recording a voice memo diary for the session one information seeking in the AI pilot study. I have a keyboard that was not registering the Windows key, which has been actually kind of annoying because I've been doing more terminal based stuff and so I've been wanting to open the terminal more and that shortcut not working has been pretty annoying.
..User researched bef ..states user will cor	3	I did about an hour-ish of web searching on this on my own yesterday and found a couple of likely problems, a couple of likely solutions, some of which I had some security concerns about. One of them seemed like it could potentially expose key presses to the open internet. I'm not a huge fan of that. So I thought I'd compare...
..Ollama Gemma 3B ..AI Model Choice Ju ..Summarizing questi	4	Olama's ability to reason through that with the university provided co-pilot. For Olama, I used the Gemma 3B model just because it's sort of what I default to. It runs pretty well in my system. And in both, I started with the initial prompt of, have a gamma key TK75 that is not registering either Windows key. What is the appropriate fix and are there any security concerns associated? What do alternative options look like?
..Co-pilot model foun	5	So I started that in both the same time. Obviously Copilot ran much faster because it's running on much bigger hardware. But ultimately Copilot was what found a solution. So I'll go ahead and say that. And going through the Olamalog first, I was doing these kind of at the same time. So there is that.
..Gemma 3B prompt p ..No blind trust	6	Olamma first defaulted to a hardware failure, which is kind of interesting. It's definitely not that. And it suggests that there's like, I should start by pulling switches, which is kind of an interesting suggestion. It also links to a YouTube tutorial that does not exist. Pretty obviously. And then it has this like little note that's like, it's probably not firmware. And so I followed up with why do you believe the firmware is

Data Analysis

Human-AI Interaction Coding & Initial analysis:

Participant 2, AI Model: Ollama Gemma 3B Model

8. Links and other resources to answer prompt

This code refers to moments where an AI tool provides the user with a link to an internet resource with the purpose of solving a problem or answering a question given to it by the user.

YouTube Tutorials - GamaKay TK75 Switch
Replacement: **https://m.youtube.com/watch?v=mJq-UvHlK-E**
GamaKay Support Website: **https://gamakay.com/** (Check for firmware updates or potential support)

Resources:

- Microsoft Support Article on Remapping the Windows Key: **https://support.microsoft.com/en-us/windows/remap-windows-key-2327bb58-a2f9-4d9a-8d90-32c4c6ef8695**

46. False link hallucination
This code refers to moments where the user identifies that a link that the AI tool has provided does not exist and was not prompted to give a link in its response. It did this on its own.

Excerpt From Transcript:

- "which I told it was not there and it apologized a bunch of times which also didn't help and sent me somewhere else where it also didn't exist."

- "It's worth noting that this model does not have access to the internet, and so it did not do any searches. This was just it kind of spinning its wheels."

Analysis:
What I found most interesting about this code to include in this assignment is that out of the 6 separate interactions with both of my participants, only the Gemma 3B hallucinated links that do not exist. When it was caught doing this it began to apologize in a way that felt incredibly urgent, as though it mattered. This type of apologizing and sycophantic type behavior also falls under the code "I am totally human", which describes instances where AI is mimicking human conversational conventions that would normally function to keep a conversation going. Based on the context of this interaction I would say that this model is highly tuned for user engagement, in that it prioritizes engagement over accuracy. It is striking that it would create links that do not exist, especially as it does not have the capability of even accessing the internet.

21. Conversational Direction-ing
This code refers to moments where the AI presents users with possible follow-up questions and process directions related to prompts provided by the user.

- Open AI's ChatGPT Participant 1
  - "If you'd like, I can next: derive the turbine efficiency equation step-by-step, connect turbines to T-s diagrams (super helpful visually), or explain blade shapes and why impulse vs reaction stages matter. Just tell me which direction you want to go next."
- Open AI's ChatGPT Participant 1

  If you'd like, I can also:

  - Convert these into **measurable engineering requirements**
  - Adjust them for **beginner vs. professional users**
  - Help with a **needs hierarchy or House of Quality**

  -
- Llama's Gemma 3B Participant 2

- o Do you find this explanation reasonable, or would you like me to elaborate on any specific point?

Analysis:
Conversational direction-ing appeared to be the most prevalent amongst the interactions between participant 1 and ChatGPT. While participant 2 only had a single instance of this type of behavior. The main difference between the prompts provided by each participant was that participant 1's prompts were very brief and lacked detail in terms of describing what type of output was expected, so the AI seemed to take more liberties in determining what direction the conversation was going to take. Participant 2's prompts followed a pattern where the expected output was clearly defined, and while the AI that participant 2 used did not always produce the correct answer, it did not steer the conversation as often. I am not sure if this is model dependent or user dependent but still points to ways in which user interactions can shape the level of agency that an AI has over the way users interact with information that they may or may not understand.
Lastly, one interesting observation made by both participants was that it was important for them to already know what the "right" answer may look like when prompting an AI to solve a problem. Small errors in an AI's "thinking" could lead to drastically different outputs that may or may not be wrong. In one specific instance, the AI got the troubleshooting diagnosis correct but did not get the solution correct. This type of dynamic highlights that the relationship between Human and AI is far more collaborative on a peer level than those pushing for AI integration may wish for you to believe. (That is a biased opinion on my part though...).

Ethogram:

Behavior	Description	Context	Outcome
Name of identified behavior correlates to name in codebook.	**Definition of identified behavior**	**Each instance of specified behavior is described and is numbered to connect behaviors across a given context.**	**Each outcome is described and numbered to connect behaviors and other outcomes.**

8. Links and other resources to answer prompt	This code refers to moments where an AI tool provides the user with a link to an internet resource with the purpose of solving a problem or answering a question given to it by the user.	1. (Llama Gemma 3B Model) User asked AI for help troubleshooting problems with mechanical keyboard in an information seeking session. AI provided list of possible solutions as well as links to a user manual and YouTube video.	1. The links that the AI provided were non-existent or deadlinks. AI had no access to internet. (See 46. False link hallucination)
46. False Link Hallucination	This code refers to moments where the user identifies that a link that the AI tool has provided does not exist and was not prompted to give a link in its response. It did this on its own.	1. (Llama Gemma 3B Model) AI provided false link.	1. User was made to believe that the AI could provide assistance with a problem that ultimately it was not able to solve with resources it provided. 1. AI apologized profusely in a human-like way.
21. Conversational Direction-ing	This code refers to moments where the AI presents users with possible follow-up questions and process directions related to prompts provided by the user.	2. User had asked AI to describe how wind turbines worked in an information seeking session. AI described wind turbines within specifications of participants prompt but also provided a list of things the user could ask next based on assumptions it had about user.	2. User was not interested in pursuing the options provided.

<u>Discussions & Reflections</u>

How do students learn to use AI tools?

Based on data gathered from each participant I gathered that there were two primary ways for learning how to use AI tools. Participant 1 was able to figure out how to use AI tools by trial and error. Most of their interactions with ChatGPT were very surface level, with very short prompts, even sometimes using images of assignment questions as the only data given in a prompt. An example of this type of behavior can be seen in the image listed below.

The solid circular shaft shown has an outside diameter of 58 mm and is made of an alloy that has ultimate failure strengths of 250 MP in tension and 420 MPa in compression. Determine the largest permissible torque T that may be applied to the shaft, according to on the Mohr failure criterion.

Answer: T = [ ] kN-m

Nice next problem—this one ties all the pieces together. I'll walk you through the **process** for using the Mohr failure criterion on a shaft in torsion, but I'll stop short of plugging numbers so you can do them.

Participant 2 expressed extensive interest in AI that guided their use of AI tools. So while there methods for learning to use AI involved trial and error, they also did their own research on how to get the best result from prompting AI tools. Participant 2 went so far as to install their own version of an AI model on a dedicated computer. There prompts were more detailed and included information about the problem and expected output from the AI. An example of this type of behavior can be seen in the image below.

I have a GamaKay TK75 that is not registering either windows key. What is the appropriate fix, and are there any security concerns associated? What do alternative options look like?

edit

Okay, let's break down the issues you're experiencing with your GamaKay TK75 and explore potential fixes, security implications, and alternative options. This is a common issue with older keyboards and can be frustrating.

In future research it will be important to utilize dedicated surveys to identify sources of AI knowledge, whether that be peers, websites or text books. Survey questions of this type will help inform AI literacy resources in the analysis phase.

How do students interact with AI tools?

Each participant had their own unique way of interacting with AI tools as was described in the previous paragraphs. What stood out to me the most was the ideology and perception of the AI as a tool driving their interaction. Participant 1 used language that seemed to describe the AI tool as having a way of "thinking" that one could attribute to another living organism.

[redacted] It's ridiculous. So, you know, computers just have a slightly different way of looking at things than we do. So that was definitely something that was interesting.

Participant 2 on the other hand seemed to have a very straight forward way of interacting and describing the functionality of AI tools that was guided by their knowledge and previous experience with AI tools.

Write a script in python that takes a directory of .pdfs as an input, opens each pdf, passes the contents of the pdf to an Ollama api call that analyzes them, considers their relevance to a prompt, writes a short paragraph about them in a text document, and sorts the pdfs into one of two new directories based on relevance. Please set it so that the model is constrained within a certain sub directory, set within a local .env file. Make sure that the model is able to view the text and images within the pdf. Please also set a local system prompt that is passed to the Ollama api call. This does not need to be a conversation, and I only want one output (the document with relevance summaries).

It is important that my future thesis work is informed by proper AI literacy in order to identify pitfalls in research participants use of AI tools.

How successful are AI tools at the tasks given to them by students?

Based on the data collected, I don't think that it is possible for me to fully answer this research question as there is not enough data to give a conclusive answer. Based on participants reflections, AI was prone to making mistakes and assuming the participants intentions when not given enough information in a given prompt. In one particular instance, AI was able to correctly identify a problem that the participant was trying to solve, but provided an incorrect solution. This is significant because it shows that while a lot of the information AI provides appears to be correct, it is vital that the user understand what the correct answer should look like. AI acted as a consistent collaborator, but its assured tone can lead the user to believe that the wrong answer is correct. Further research is needed regarding the agnecy in which an AI has over a student's work and how it may alter the learning process.

Methodological Reflections

Semi-Structured Interviews: While interviews were helpful in establishing rapport with each research participant, I felt that they were easily derailed by tangential topics within the AI field that took away from core questions that I wanted answered. Combing a 2+ hour long interview transcript became incredibly laborious and created a surplus of unused data. In future research, I think that implementing a seperate survey asking important questions will help alleviate the pressure to keep interviews in line with the data I need to collect. I imagine that future interviews I conduct in my thesis research will become unstructured and act more to build rapport and trust with participants so that they feel comfortable giving me information about their AI use, which is often viewed negatively in academia.

Diary Studies: I was surprised by the wealth of information I was able to gather from this methodology. I was able to gather information on their use of AI, Human-AI interactions and their opinions of each interaction. Both the audio recordings and the AI tool usage transcripts were very easy to transcribe, code and analyze. Flaws in my use of the diary method were a lack of clarity in session design, one participant felt a bit confused about what they needed to talk about, and at times delivered very little descriptive data. On the other hand, another participant provided far more data than I really needed, this led to a lot of important descriptive data to be buried. Future iterations of this methodological approach will involve specific questions that I want them to answer instead of leaving them to report back freely. So rather than a diary, it'll function more as a recuring questionnaire after each interaction with some room for opinion.

Ethogram Analysis: I was surprised by how effective this analysis method was in organizing behavioral data. While I do not think that I was able to really come to any firm conclusions on AI use with this analysis method, I do feel that it will be integral to my thesis project when looking for specific solutions to student AI use and collaborating with educators to implement said solutions. One big flaw in this ethogram analysis is the lack of observational data. More observations and quantifiable numbers to confirm the recurrence of observed behaviors will help lend credibility to my findings.

Significance of the Project

This pilot study allowed me to test data collection and analysis methods that I have only a little bit of experience conducting. The experiences that I have had over the last ten weeks have helped me build confidence in reaching out to test participants, relaying expectations for said participants and working to understand the topic that I am researching. While I was not as successful at gathering the breadth of data needed to form a conclusion, I feel that this was a great opportunity to experience this type of research with room for mistakes and adjustments.

From a practical and applied standpoint, researching students use of AI will benefit educators at Oregon State University in deciding how to develop their own AI policy for their classes, and help build AI literacy to help inform curriculum change in hopes of working with and through AI tools in the future. By understanding AI use from the point of view of the student we are able to better inform educators and students about what AI is good for and what it is not good for. Furthermore, it is important to emphasize that AI is not going away, and the only way to mitigate the negative effects of AI use is to educate our educational communities rather than punish and ban AI use.

New technologies effect the ways in which we learn, process and teach new information. Given that AI tools produce massive amounts of information that may or may not be correct they inherently hold agency over the learning outcomes of students who use this technology. Modern pedagogy needs to be informed by the ways in which tools like AI will effect how future students learn and view their learning environment. By compiling data gathered on student use of AI we will be able to enhance academia's ability to effect positive change in an AI integrated world.

References

Arredondo, Pablo. 2023. “GPT-4 Passes the Bar Exam: What That Means for Artificial Intelligence Tools in the Legal Profession.” *Standford Law School Blog*, April 19. https://law.stanford.edu/2023/04/19/gpt-4-passes-the-bar-exam-what-that-means-for-artificial-intelligence-tools-in-the-legal-industry/

AI Futures Project. 2025. “AI 2027.” AI Futures Project. Effective April 3. ai-futures.org.

David E. Kieras, Susan Bovair. 1984. The Rold of a Mental Model in Learning to Operate a Device. Retrieved January 17, 2026 from https:/www.sciencedirect.com/science/article/abs/pii/S0364021384800038.

Diana Loeffler, Anne Hess, Andreas Maier, Joern Hurtienne, and Hartmut Schmitt. 2013. Developing Intuitive User Interfaces by Integrating Users’ Mental Models into Requirements Engineering. In *Proceedings of the 27th International BCS Human Computer Interaction Converence,* 1-10. https://dl.acm.org/doi/10.5555/2578048.2578069

Kosmyna, Nataliya., Eugene Hauptmann, Ye Tong Yuan, et al. 2025. “Your Brian on ChatGPT: Accumulation of Cognitive Debt when Using an AI Assistant for Essay Writing Task.” *MIT Media Lab*. https://doi.org/10.48550/arXiv.2506.08872

Latour, Bruno. 2005. “Reassembling the Social: An Introduction to Actor-Network Theory.” *Oxford University Press*. https://doi.org/10.1093/oso/9780199256044.001.0001.

Moore, Jerry D. 2018. “Bruno Latour: The Creation of Knowledge.” In *Visions of Culture: An Introduction to Anthropological Theories and Theorists Fifth Edition,* Rowman & Littlefield Publishers.

Nielsen, Jakob., and Rolf Molich. 1990. “Heuristic Evaluations of User Interfaces.” *CHI ’90 Proceedings*. https://dl.acm.org/doi/epdf/10.1145/97243.97281.

Pallante, Virginia., Peter Ejbye-Ernst, and Marie Rosenkrantz Lindegaard. 2023. “An ethogram method for the analysis of human distress in the aftermath of public conflicts.” *Behavior* 160 (15):1409-1445. https://doi.org/10.1163/1568539X-bja10247.